VULNERABILITY ASSESSMENT OF HONOLULU'S WATER SUPPLY

The **Impacts of Climate Change on Honolulu Water Supplies and Planning Strategies for Mitigation** report, commissioned by the Honolulu Board of Water Supply, assesses how projected changes in climate have made the city's freshwater supply system increasingly vulnerable. Honolulu relies almost entirely on groundwater, making it highly sensitive to long-term rising temperatures, sea level rise (SLR), and shifts in weather patterns. This assessment uses regional climate projections and hydrologic models to evaluate how these stressors may affect aquifer recharge rates, water infrastructure, water intrusion, and future water demand.

CLIMATE CHANGE IMPACTS

SEA LEVEL RISE

Sea levels are projected to rise by 1 ft by 2050 and 3.2 ft by 2100. This impact will cause increased coastal erosion, groundwater inundation, and marine inundation for Honolulu.

COASTAL EROSION

- Highest risk
- Saltwater intrusion into aquifer
- Damage to roadways
- Pipe corrosion

GROUNDWATER INUNDATION

- Localized flooding from below
- Risk of saltwater intrusion into aquifers
- Rise of the groundwater table
- Pipe corrosion and main breaks

MARINE INUNDATION

- Pipeline corrosion
- Flooding of infrastructure
- Submergence of roadways and bridges

EXTREME WEATHER EVENTS

Extreme weather events are expected to become more frequent and severe with climate change. This impact will cause more intense rainfall and long-term declines in precipitation

HEAVY RAINFALL

- Increased surface runoff causes less infiltration of water into the ground
- Recharge efficiency decreases with higher rainfall intensity
- Results in lower aquifer recharge

DECLINE IN RAINFALL

- Decreases future recharge rates of aquifers
- Decreases the sustainable yield of freshwater extraction
- Overall decrease in available freshwater supply

EXPOSURE



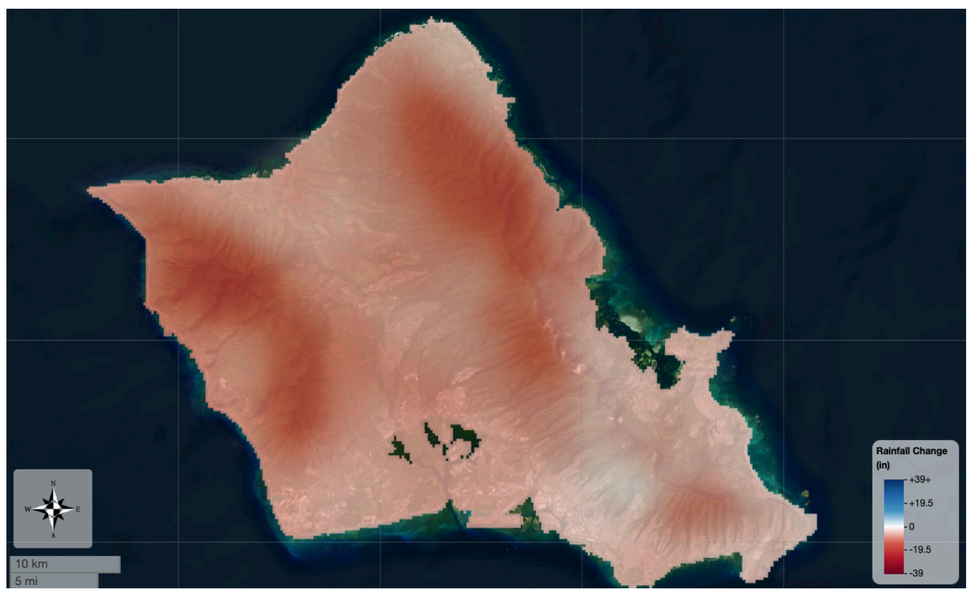
Coastal Erosion (Blue) with 3.2 ft of SLR

The left map (1) depicts future areas of coastal erosion with 3.2 feet of sea level rise in 2100. The assessment highlights that coastal erosion will have the highest risk due to saltwater intrusion and damage to infrastructure. Many of Honolulu's beaches will be susceptible to coastal erosion (shown in blue), depicting the areas at risk of intrusion and infrastructure damage.



Groundwater (Green) & Marine Inundation (Blue) with 4 ft. of SLR

The left map (2) depicts future areas of Groundwater and Marine Inundation with four feet of sea level rise. The assessment states that these inundations will also cause intrusion and damage to infrastructure. Marine Inundation (shown in blue) and Groundwater Inundation (shown in green) further depict areas at risk of intrusion and infrastructure damage.



Statistically Downscaled Annual Rainfall (RCP 4.5) - 2040-2069

The above map (3) depicts the statistically downscaled climate projections for the island of O'ahu under RCP 4.5 emissions scenario, focusing on the mid-21st century. The assessment highlights that changes in precipitation for Honolulu range from -11% during the wet season and -21 to -28% during the dry season. These shifts lead to significant reductions in groundwater replenishment, decreasing future recharge rates and putting stress on Honolulu's long-term freshwater supply and aquifer sustainability

1. Map sourced from O'ahu Climate Web Explorer, similar map included in report as figure 4-8
2. Map sourced from NOAA, included in report as figure 5-10
3. Map sourced from Hawaii Climate Data Portal, similar map included in report as figure 2-8

SOLUTIONS

The report claims the following solutions to impacts of sea level rise and extreme weather events in these three sectors:

INFRASTRUCTURE RESILIENCE

- Shift groundwater pumping to inland and upslope wells
- Plan for well replacement and infrastructure relocation

WATER SUPPLY

- Investigate and manage aquifer recharge and storage
- Implement water shortage plans with clear curtailment stages
- Increase storage capacity to buffer against dry periods
- Expand water conservation efforts through green plumbing codes
- Capture and reuse stormwater & greywater at homes and public facilities

WATER QUALITY

- Monitor and manage the saltwater-freshwater interface
- Incorporate green infrastructure to increase water filtration

CRP 3545 / FA 2025 / Tomás Schmieder & Ali Hartung

ASSESSMENT

STRENGTHS

The report draws on regional climate projections and hydrological models to estimate changes in rainfall patterns, groundwater recharge, and sea level rise. There is a clear focus on physical infrastructure and water supply reliability, providing a detailed analysis of factors that affect the vulnerability of key infrastructure in coastal areas.

WEAKNESSES

The assessment does not consider how climate-related water threats may impact different communities, with no discussion of social vulnerability, equity, or affordability. Honolulu has a large tourist culture; the report avoids discussing the disparity of local and resort water use. Moreover, while infrastructure exposure is mapped, the report does not examine who is most affected by service disruption or which populations may have the least adaptive capacity. Key climate drivers such as drought and rising temperatures are underrepresented, with no analysis of how heat waves could exacerbate water stress and overuse. Although the assessment provides potential strategies, it lacks specificity about who is responsible for implementing them, what funding is required, or community engagement.

MODIFICATIONS

Future assessments need to evaluate hydraulic modeling in combination with social vulnerability assessments to accurately understand the future of Honolulu's water supply. Infrastructure vulnerability affects peoples' access to freshwater. Therefore, reports must identify which communities are most at risk. Once determined, recommendations should include a local perspective to understand possible implementations and unintended consequences. Assessments need to include further information on scenarios of drought, rising temperatures, and their impact. This report needs to incorporate the cost of damages in its initial vulnerability assessment. Similarly, it needs to analyze the cost of solutions and possible sources of funding. These modifications hope to create a more realistic, well-rounded assessment on the future vulnerability of Honolulu's water supply.